

CSTE 5131

Machine Learning

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Classification: Basic Concepts

- Classification problem overview
- General issues of classification DM
- Mining classification model by decision tree induction
- Bayesian Classification
- Other classification methods
- Summary

Other Classification Methods

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Logistic Regression



K-Nearest Neighbor (KNN)

- K-Nearest Neighbors (KNN) method is a simple but very effective method for classification
- The classifiers do not use any model to fit and only based on memory
- How does it work?
 - Determine parameter k , the number of nearest neighbors
 - Calculate the distances between the query instance and all the training samples by using the standard Euclidean Distance.

$$\text{dist}(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_{s=1}^m (x_s - x'_s)^2}$$



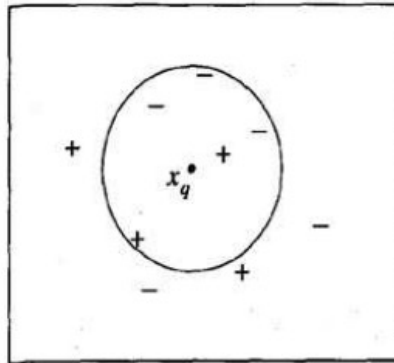
K-Nearest Neighbor (KNN)

- How does it work? (Cont.)
 - Determine the k nearest neighbours based on the distances and gather the class labels of the k nearest neighbors.
 - For discrete-valued target functions, assign the label with the majority in the k nearest neighbors as the predicted class label of the query instance.
 - For continuous-valued target functions, compute the mean value of k nearest training examples.



KNN Example

- The following figure illustrates KNN algorithm, where the target function is boolean-valued



- 1-nearest neighbour, x_q is classified as '+'
- 5-nearest neighbours, x_q is classified as '-'

- Classifying

Drawbacks of KNN

- 
- The fixed value of k
 - All k neighbours have an equal contribution to predict a target class
 - Choice of metric
 - Difficulty of handling unbalanced Data

• Classifying

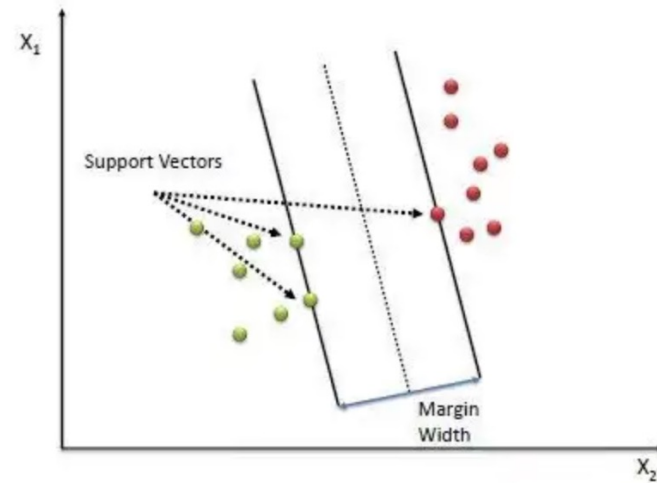
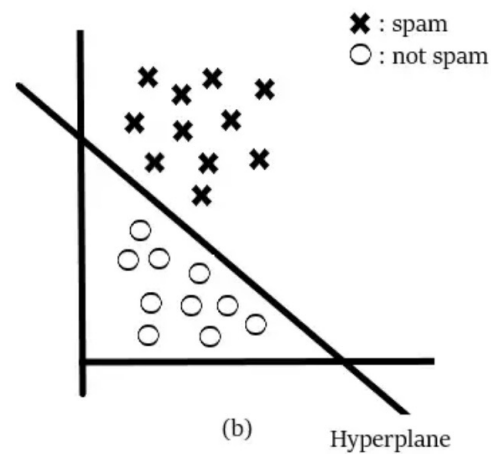
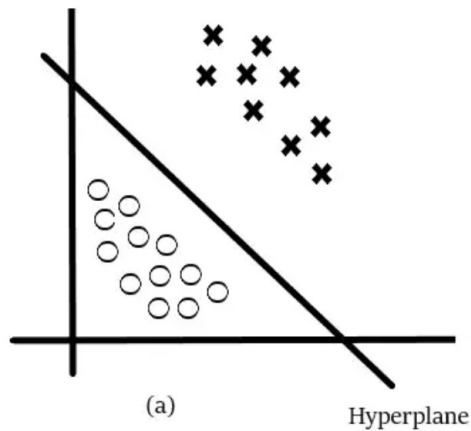


Support Vector Machine (SVM)

- Support Vector Machines (SVMs) (Cortes & Vapnik 1995) can efficiently perform both linear and non-linear classifications.
- For linear support vector machines, SVMs construct a maximum margin hyperplane, a decision boundary that divides the observations into two classes (Russell & Norvig 2009).
- The support vectors are formed by the closest points from the hyperplane.
- SVMs are very robust to noise in the data set and can avoid the loss of generalization if the parameters for a given kernel function are chosen appropriately (Auria & Moro 2008). However, there is not a universal selection criterion for choosing which kernel function can be used to best classify data.



Support Vector Machine (SVM)



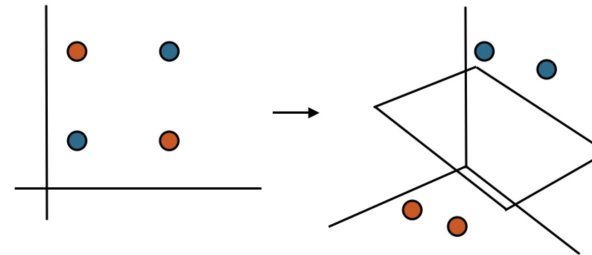
<https://towardsdatascience.com/support-vector-machines-svm-c9ef22815589>



Support Vector Machine (SVM)

- Terminologies used in SVM:
 - ❖ Hyperplane
 - ❖ Support vector points
 - ❖ Margins
 - ❖ Kernel trick
 - linear kernel
 - polynomial kernel
 - Radial basis function kernel (RBF)/
Gaussian Kernel

Kernel function implicitly maps data from its original space to a higher dimensional feature space.





Logistic Regression

- **Logistic Regression** models the probabilities of the different possible outcomes of a response variable, given a set of input variables. The model is appropriate for two-class classification or prediction problem.
- The logistic regression classifier can be derived by analogy to the **linear regression hypothesis**.
- Regression is accomplished by using a linear regression algorithm, and classification is achieved through logistic regression.
 - Using Linear Regression we can form (equation for the best-fitted line)

$$Y = mx + c$$

- Using Logistic regression,

$$Y = \text{logistic}(mx + c)$$

Sigmoid function $\sigma(x) = \frac{1}{1 + e^{-x}}$



Homework!

- Support vector machine:

<https://towardsdatascience.com/support-vector-classifier-explained-a-visual-guide-with-mini-2d-dataset-62e831e7b9e9/>

- Logistic regression:

<https://towardsdatascience.com/a-handbook-for-logistic-regression-bb2d0dc6d8a8>

Thank you

